

Assignment: Module - 5 Network Fundamentals and Building Networks

Section 1: Multiple Choice

1.) What is the primary function of a router in a computer network?

Ans-1(C) Forwarding data packets between networks

2.) What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?

Ans-2 (D) Dynamically assigning IP addresses to devices

3.) Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?

Ans-3 (B) Switch

4.) Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?

Ans-4 (B) Bus

Section 2: True or false

5.) True or False: A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

Ans-5 True

6.) True or False: TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

Ans-6 False

TCP is a connection-oriented protocol, not connectionless. It ensures reliable, ordered, and error-checked delivery of data.

7.) True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Ans-7 True

Section 3 : Short Answer

8.) Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Ans-8 Setting up a wireless network for a Small Office/Home Office (SOHO) involves several essential steps to ensure secure, reliable, and efficient connectivity for all devices. Here's a detailed breakdown of the process:

1. Select the Right Wireless Router or Access Point (AP)

- **Choose a wireless router:** Ensure it supports the latest Wi-Fi standards, such as Wi-Fi 5 (802.11ac) or Wi-Fi 6 (802.11ax), depending on your needs. Wi-Fi 6 offers better speed, coverage, and efficiency.
- **Check for additional features:** Features like Quality of Service (QoS), firewall protection, VPN support, and dual-band capability (2.4 GHz and 5 GHz) are important for optimal performance.

2. Physically Set Up the Router

- **Position the router:** Place the router in a central location in the office or home for the best coverage. Avoid placing it near metal objects, microwaves, or thick walls that can cause interference.
- **Connect to the modem:** If you have a separate modem, plug the router's WAN (Internet) port into the modem using an Ethernet cable.

3. Power On the Router

- **Turn on the router:** After ensuring everything is connected, power on the router. Wait for the status lights to show that it's ready to use.

4. Access the Router's Web Interface

- **Connect to the network:** Use an Ethernet cable to connect your computer to the router's LAN port or connect wirelessly using the default Wi-Fi settings (SSID and password found on the router label).
- **Log in to the router:** Open a web browser and enter the router's default IP address (typically **192.168.1.1** or **192.168.0.1**). Use the default username and password, often printed on the router label or in the manual.

5. Configure Basic Router Settings

- **Change the admin login:** For security, change the default admin username and password to something unique.
- **Set up Internet connection settings:** If necessary, configure the router to connect to your ISP (Internet Service Provider) by entering your PPPoE, DHCP, or static IP details.

6. Configure Wireless Settings

- **Change the SSID (network name):** Set a unique name for your Wi-Fi network. This helps differentiate it from other nearby networks.
- **Set up Wi-Fi security:** Enable WPA2 or WPA3 encryption to secure the network. Choose a strong password (at least 12 characters) to prevent unauthorized access.
- **Select the wireless channel:** If possible, choose a channel with less interference. Routers often automatically choose the best channel, but you can manually select one if needed.
- **Choose dual-band or separate networks:** If your router supports dual-band (2.4 GHz and 5 GHz), you can either create two separate SSIDs or allow devices to automatically choose the best band.

7. Enable DHCP for IP Address Assignment

- **Configure DHCP:** Ensure that the router's DHCP feature is enabled, which will automatically assign IP addresses to devices on the network. Set the DHCP range to accommodate the number of devices that will connect to the network.

8. Set Up a Guest Network (Optional)

- **Create a guest Wi-Fi network:** For visitors or clients who need Internet access but shouldn't have access to your internal network, set up a separate guest network with its own SSID and security settings.
- **Enable isolation:** Make sure the guest network is isolated from your main network, preventing guests from accessing sensitive files or devices.

9. Test the Network

- **Connect devices:** Use various devices (computers, smartphones, printers) to connect to the wireless network and check the signal strength and speed.
- **Test coverage:** Move around the office or home to ensure that the signal is strong in all areas. If necessary, reposition the router to improve coverage.

10. Secure the Network

- **Change default settings:** Update default credentials, such as admin username/password and Wi-Fi credentials.
- **Enable firewall:** Ensure the router's built-in firewall is activated to protect your network from external threats.
- **Disable WPS:** If not needed, disable Wi-Fi Protected Setup (WPS), as it may pose a security risk.

11. Update Router Firmware

- **Check for firmware updates:** Visit the router manufacturer's website or use the router's interface to check for firmware updates. Keeping your router's firmware up-to-date ensures better security and performance.

12. Monitor and Troubleshoot the Network

- **Monitor network traffic:** Most routers have a dashboard where you can see connected devices and data usage.
- **Troubleshoot issues:** If there are any connectivity problems, try restarting the router, checking for interference, or updating firmware.

13. Backup Configuration (Optional)

- **Save settings:** After the network is set up, you can back up the configuration of the router. This allows you to restore the settings in case of a reset or router replacement.

Section 4: Practical

9.) Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol)

Ans-9 Done

Section 5: Essay

10.) Discuss the importance of network documentation in the context of building and managing networks.

Ans-10 Network documentation is an essential practice in building and managing networks. It involves recording details about the network's architecture, components, configurations, and policies.

- ❖ **The importance of network documentation can be discussed in several key areas:**

1.) Improved Troubleshooting and Maintenance:

- **Quick identification of issues:** Detailed network documentation allows IT staff to quickly identify and diagnose problems. It provides information about network components, IP addressing schemes, device configurations, and interconnections, which is vital for troubleshooting issues such as connectivity failures or performance bottlenecks.
- **Efficient root cause analysis:** When issues arise, network documentation helps identify potential causes by giving network administrators access to configuration settings, network topology, and change history. This makes finding the root cause much faster than relying on memory alone.

2.) Network Planning and Expansion:

- **Scaling the network:** Network documentation aids in planning for future growth. By understanding the existing network layout and capacities (such as IP address usage, available bandwidth, and device configurations), it's easier to plan for network expansion without causing disruptions.
- **Avoiding network conflicts:** Documentation ensures that future additions or changes to the network won't conflict with existing infrastructure. It helps prevent issues like IP address conflicts or incorrect configuration of switches, routers, or firewalls.

3.) Security and Compliance:

- **Risk management:** Clear and accurate documentation helps identify vulnerabilities in the network. It ensures that security devices (firewalls, intrusion detection systems, etc.) are properly configured, and that there are no gaps in security policies.
- **Compliance with regulations:** Many industries require strict compliance with security and operational standards. Network documentation can help demonstrate that an organization is adhering to these regulations, such as PCI-DSS, HIPAA, or GDPR, by showing that network security protocols, access controls, and audit trails are in place.

4.) Facilitating Troubleshooting and Collaboration:

- **Onboarding new team members:** Network documentation is essential when bringing new team members up to speed. It provides a clear understanding of

the network's design, configuration, and components, making it easier for new employees to familiarize themselves with the infrastructure.

- **Collaborative decision-making:** When multiple teams are involved in managing the network (e.g., IT, security, development), network documentation serves as a common reference, ensuring everyone is on the same page regarding the network's structure, security policies, and configurations.

5.) Disaster Recovery and Business Continuity:

- **Backup and recovery planning:** In the event of a network failure, disaster recovery becomes much more efficient if the network documentation is up-to-date. Detailed records about network configurations, hardware, and connections allow the recovery process to be faster and more accurate, reducing downtime and preventing data loss.
- **Minimizing impact of network outages:** Knowing exactly how the network is designed, which components are redundant, and how to restore services is crucial for minimizing business disruption during incidents.

6.) Regulatory and Vendor Support:

- **Vendor support:** When dealing with vendor support, accurate documentation can be crucial in helping vendors understand the environment to provide better support or resolve issues more quickly. This includes information about device models, firmware versions, and configurations.
- **Audit trails:** In case of audits, either from internal teams or external regulatory bodies, having up-to-date documentation ensures that the network complies with required standards and regulations, and that changes are tracked and documented for accountability.

7.) Optimizing Network Performance:

- **Capacity planning:** By documenting bandwidth usage, device performance, and other network metrics, it becomes easier to identify performance bottlenecks and optimize resource allocation. This can help ensure the network runs at peak efficiency.
- **Preventing over-provisioning/under-provisioning:** Accurate network documentation helps avoid unnecessary expenses, such as over-provisioning resources, by providing visibility into actual usage trends. It also helps avoid under-provisioning, which could lead to performance degradation or service interruptions.

8.) Change Management:

- **Tracking network changes:** Network documentation serves as a record of all changes made to the network, including configuration updates, new devices added, and software upgrades.
- **Reverting changes:** If a change causes issues, network documentation allows administrators to quickly revert back to a previous configuration by providing the necessary details about the network's prior state.

9.) Standardization:

- **Consistency across the network:** Documentation helps ensure that the network is built and maintained in a consistent manner. Standard naming conventions, IP address schemes, and device configurations are easier to enforce when properly documented.
- **Streamlining network operations:** By following established guidelines and documented best practices, network administrators can streamline their operations, reduce mistakes, and increase efficiency.